- All problems are worth an equal weight, even if some may be harder than others.
- Write answers on a separate sheet, either typed or handwritten, either on paper or a note taking device. No need to copy the problem statements. Please, upload an unique pdf file to Gradescope and assign each question to its corresponding page.
- Late submissions will be worth significantly less points.
- Circle your final answer, but make sure to include the work to get there.
- It is OK to work with others, as long as you acknowledge them in your paper and write down your solutions in your own words. It is not OK to copy a solution from whatever source.

1. Use an integrating factor to solve the following equations:

a) $y' = \frac{4}{t}y + \frac{1}{t^3}$.

b) $y' = \frac{4t^3 + 4t}{t^4 + 2t^2}y + t^4 - 2t^2$.

c) $y' = -5y + 3\sin t$.

Remark: This is an example of when using undetermined coefficients would have been easier.

2. Use undetermined coefficients to solve the following equations:

a) $y' = 3y + t^2 + 2$.

b) $y' = -\frac{y}{3} + \cos 2t$.

c) $y' = y + 2t\sin 3t$.

Hint: For a particular solution here, try one of the form

$$y_P(t) = A\sin 3t + B\cos 3t + Ct\sin 3t + Dt\cos 3t$$

3. In class, we have learned that, when solving $\alpha y'' + \beta y' + \gamma y = 0$, if there is only one root λ to the characteristic equation $\alpha\lambda^2 + \beta\lambda + \gamma = 0$, then the two independent solutions to use are $y_1(t) = e^{\lambda t}$ and $y_2(t) = te^{\lambda t}$, and the general solution is $C_1e^{\lambda t} + C_2te^{\lambda t}$. We also verified that y_1 is indeed a solution of the differential equation, but we haven't verified it for y_2. **Verify by plugging-in that $\mathbf{y_2}$ is a solution to the DE in this case.**
Hint: Find a formula for λ in terms of α, β, γ, and you'll need to use the fact that $\beta^2 - 4\alpha\gamma = 0$, which is what says that there is only one root of the characteristic equation.

4. An industrial container has a capacity of 144L, but initially it contains only 9L of water. Dissolved in this water are initially 10g of sugar. Water containing $\frac{2}{3}$g/L of sugar is pouring into it at a rate of 9L/min, but due to a hole in the container the contents are leaking at a rate of 3L/min. Assume the sugar is spread out evenly through the water in the container at all times. How many grams of sugar will there be in the container by the time when it's full?
Hint: Notice this is a problem in which the volume of the contents does not stay constant.

5. An object of mass 1 attached to the end of a horizontal spring on a table is initially 4 units of distance to the left of the equilibrium position, but moving with speed 13 to the right. The spring is anchored to a wall on the left, as usual. The spring constant is 6 and the friction coefficient of the table is 5. What distance to the right of the equilibrium point will the object reach before the spring starts pulling it back in?

6. Consider a harmonic oscillator with parameters $m = 1$, $k = 2$, and $b = 1$. This gives rise to underdamped motion, so there will be periodic oscillations.

a) What is the frequency of these oscillations?

b) If the object on the end of the spring was just a little bit heavier, would this frequency increase or decrease?

c) If the spring was just a little bit stronger (larger k), would this frequency increase or decrease?

d) If the object experienced just a little bit more friction (larger b), would this frequency increase or decrease?

*Note: The **frequency** of a function given as a combination of terms in the form* $\sin(\omega t)$ *and* $\cos(\omega t)$ *is defined as* $\nu = \omega/2\pi$.

7. Verify that the functions $x(t) = \cos 2t + 3\sin 2t$ and $y(t) = \cos 2t + 5\sin 2t$ are solutions to the initial value system

$$\begin{cases} x' = 16x - 10y \\ y' = 26x - 16y \\ x(\pi/4) = 3\ ,\ y(\pi/4) = 5 \end{cases}$$

8. Answer the following items about predator-prey systems:

a) In the following system, the constants A, B, C, D, N are positive. Which function, f or g, represents the predators and which represents the prey? How does the prey population behave in the absence of predators?

$$\begin{cases} f' = -Af + Bfg \\ g' = Cg\left(1 - \dfrac{g}{N}\right) - Dfg \end{cases}$$

b) Suppose three animal species, X, Y and Z, live in a preserve, and let x, y, z denote the number of individuals of each, in that order. Suppose X is a predator species that preys on both Y and Z, while these two are species that cooperate among themselves and thrive in the presence of each other. Write down a possible model system of differential equations for x, y and z.

9. Find the solution to the following IVS: For future reference, change initial values to $t = 1$ and easier constants.

$$\begin{cases} x' = \dfrac{2(x-3)}{t} \\ y' = 2y + x \\ z' = 2z - y \\ x(0) = y(0) = z(0) = 3 \end{cases}$$

10. Consider the differential system This exercise has a curve of equilibrium points, if you haven't given an example of this in class, avoid it

$$\begin{cases} x' = y^3 + 2xy - y \\ y' = x - xy^2 - 2x^2 \end{cases}$$

a) Find its equilibrium points.

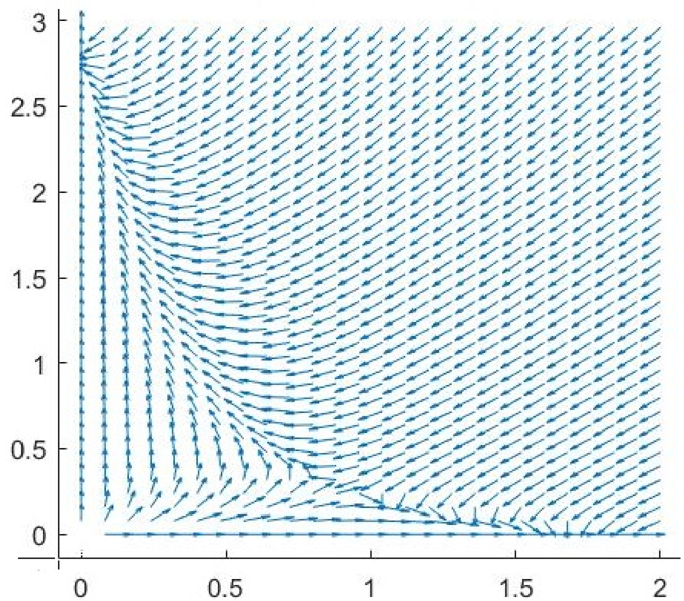
b) Find a function $H(x, y)$ with the property that all solutions to the system satisfy $H(x(t), y(t)) = \text{const.}$ as t varies.

Hint: Supplement N3 might help.

11. The following system models two competing populations living in the same space: left alone, each follows their own logistic model, but interactions between them cause both populations to go down.

$$\begin{cases} x' = x\left(1 - \dfrac{x}{2}\right) - 2xy \\ y' = 3y\left(1 - \dfrac{y}{3}\right) - 3xy \end{cases}$$

The picture on the next page is a sketch of the direction field for the system:



a) Find all equilibrium points. Notice that the direction field tells you approximately where they are.

b) Use the direction field to sketch the graphs of $x(t)$ and $y(t)$ if the initial populations are $x(0) = 2$ and $y(0) = 1.5$.

c) Use the direction field to sketch the graphs of $x(t)$ and $y(t)$ if the initial populations are $x(0) = 2$ and $y(0) = 1$.

12. Consider the initial value system

$$\begin{cases} x' = 3y - x \\ y' = 3x - y \\ x(0) = -1/2 \ , \ y(0) = 1 \end{cases}$$

a) Obtain a sketch of the differential equations' direction field. Draw a neat, big picture with clearly defined arrows spanning the entire region drawn! *Note that the initial point given has no significance for this part.*

b) Compute 3 steps of Euler's method using $\Delta t = 0.5$ and the initial value given in the IVS. Plot the results on the plane.

c) Compute 5 steps of Euler's method using $\Delta t = 0.1$ and the initial value given in the IVS. Plot the results on the plane.

d) Based on your direction field, which plot (the one from (b) or the one from (c)) is more accurate?

Calculators allowed.

13. Consider the following 3rd order IVP:

$$\begin{cases} y''' + (y')^2 - ty^3 = 0 \\ y(0) = 2\ ,\ y'(0) = 3\ ,\ y''(0) = 0 \end{cases}$$

Convert it into an initial value system of three 1st order equations, and use Euler's method with $\Delta t = 0.25$ to find an approximate value for $y(1)$.

Calculators allowed.

14. Consider the linear, non-homogeneous, autonomous system

$$\begin{cases} x' = y + 2 \\ y' = 2x + y - 4 \end{cases}$$

a) What is it that makes it non-homogeneous and autonomous?

b) Verify that the following are solutions of it:

$$\begin{cases} x_A(t) = 3 \\ y_A(t) = -2 \end{cases} \qquad \begin{cases} x_B(t) = 3 + e^{2t} \\ y_B(t) = -2 + 2e^{2t} \end{cases} \qquad \begin{cases} x_C(t) = 3 - e^{-t} \\ y_C(t) = -2 + e^{-t} \end{cases}$$

c) Draw the solution curves on the phase plane corresponding to each of these solutions.

d) What does the existence and uniqueness theorem for systems allow us to conclude about the particular solution that starts at $x(0) = 0$, $y(0) = 1000$?

15. Consider the 2nd order equation

$$y'' - 4y' + 85y = 0$$

a) Find two solutions $y_1(t)$ and $y_2(t)$ such that the general solution can be written as

$$y(t) = C_1 y_1(t) + C_2 y_2(t)$$

b) Convert the equation into a system of two 1st order equations. Obtain two solution vectors from the y_1 and y_2 that you found above. Verify that they are independent solutions. Use them to write the general solution of the system.

Notice that this approach gives us a way to define what we mean by two ***independent*** *solutions to a* 2^{nd} *order equation. It means that the vectors obtained from them are linearly independent at every point.*